How-To Library

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Sizing recommendation and performance optimization for Metadata Command Center

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Abstract

This article is intended to provide general guidelines about sizing recommendations and performance improvements for metadata extraction and profiling jobs that run on secure agents in Metadata Command Center. This article includes sizing information based only on Oracle and Amazon S3 catalog sources.

Supported Versions

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Recommendations to optimize performance

Here are some recommendations to help you optimize the performance of your tasks in Metadata Command Center:

- If you're using a secure agent-based runtime environment, consider using a secure agent in the same cloud ecosystem as the catalog source. This reduces network latency and optimizes performance. For instance, use an AWS-based secure agent to extract metadata from an Amazon S3 catalog source.
- For lower asset load, such as 10,000 assets, you can run the data profiling task on a single secure agent. For higher asset load, such as 100,000 assets, consider adding two or four secure agents to significantly reduce the run time and linearly improve the profiling performance.

Overview

Performance and sizing depends on the volume of data that Metadata Command Center processes and the runtime environment that you deploy for running the tasks.

Based on the size of data and the runtime environment, you must add additional memory and CPU cores to optimize the performance of tasks that you run in Metadata Command Center. You must also note the minimum number of cores that are required to deploy supported data sizes.

Sizing recommendations for metadata extraction tasks

You can extract metadata from external sources, such as databases, data warehouses, business glossaries, data integration resources, or business intelligence reports. The size of the metadata depends on the number of assets that are processed. Assets include the sum of databases, schemas, tables, files, columns, reports, mappings, and so on in the deployment environment. The performance of metadata extraction tasks in Metadata Command Center depends on the number of assets that are processed.

Oracle catalog source

The following table suggests sizing recommendations for extracting up to 100,000 assets (for example, 1000 tables with 100 columns per table) from an Oracle source system when the metadata extraction task runs on a Secure Agent:

Spec	Recommendation
CPU Cores	16
Memory	64 GB
Disk Size or Storage	200 GB
Network	Bonded Gigabit Ethernet or 10 Gigabit Ethernet between the source system and Secure Agent.

Amazon S3 catalog source

The following table suggests sizing recommendations for extracting up to 225,000 assets from an Amazon S3 source system when the metadata extraction task runs on a Secure Agent:

Spec	Recommendation
CPU Cores	16
Memory	64 GB
Disk Size or Storage	200 GB
Network	Bonded Gigabit Ethernet or 10 Gigabit Ethernet between the source system and Secure Agent.

Note: The extracted assets may include a combination of objects, such as folders, Parquet, AVRO, and CSV files.

Sizing recommendations for data profiling tasks

You can optimize data profiling tasks and reduce the profiling run time by configuring properties for the Data Integration Service and Metadata Platform Service in Administrator.

Data Integration Service recommendations

To optimize data profiling and reduce the profiling run time, go to **Runtime Environments** in Informatica Intelligent Cloud Services Administrator. Select the Secure Agent and set the `maxDTMProcesses` Data Integration Service property to `12`.

The following table suggests sizing recommendations for profiling up to 50,000 assets that includes sampling for two million rows for an Oracle catalog source when the data profiling task runs on a Secure Agent:

Spec	Recommendation
CPU Cores	16
Memory	64 GB

Spec	Recommendation
Disk Size or Storage	200 GB
Network	Bonded Gigabit Ethernet or 10 Gigabit Ethernet between the source system and Secure Agent.

Calculate maxDTMProcesses value

You can calculate the recommended `maxDTMProcesses` value based on the total number of CPU cores available in a Secure Agent.

The recommended `maxDTMProcesses` value is 0.75 * total CPU cores. The Data Transformation Manager (DTM) process is an operating system process that the Data Integration Service starts to run DTM instances. Based on how you configure the Data Integration Service, the service can run each DTM instance in a separate DTM process on the local or on a remote node.

For example, if the total CPU cores available in a Secure Agent is 16, then the recommended `maxDTMProcesses` value is calculated as follows 0.75*16=12. When 75% of the total CPU cores is configured to run the DTM Process, you can observe that the DTM Process utilizes 80% of the CPU cores.

Note: When you run data profiling tasks by utilizing 75% of the total CPU cores, Informatica recommends to hold other catalog source runs.

Metadata Platform Service recommendations

To optimize and reduce the failure rate of data profiling tasks, you can configure the following Metadata Platform Service properties in Informatica Intelligent Cloud Services Administrator:

- `mps_maxDeploymentBatchSize`
- `mps_maxParallelTasks`
- `mps_metadataComputeProfile`

The following table suggests the Metadata Platform Service configurations for profiling of 500 tables, where each table includes 190 columns and 100000 rows for an Oracle catalog source when the data profiling task runs on a Secure Agent:

Type	Name	Description	Recommended value	Default value
APP_CFG	mps_maxDeploymentBatchSize	The maximum number of objects that a batch profiling sub-task can contain.	50	100
APP_CFG	mps_maxParallelTasks	The maximum number of batch profiling sub-tasks that can run at the same time.	4	4
APP_CFG	mps_metadataComputeProfile	Defines the load type.	'Small:core.DataElement=1000;Medium:core.DataElement=10000;'	'Small:core.DataElement=1000;Medium:core.DataElement=10000;'

Sizing Recommendations

To tune the performance, configure the properties of the Data Integration Service and Metadata Platform Service based on the size of data. The minimum recommended hardware sizing requirement to profile 3.5 K tables with 350 K columns is 6 CPU cores and 6 GB of memory.

The following table suggests the configurations to profile an Oracle catalog source when the data profiling task runs on a Secure Agent:

Table 1. 5 K tables and 500 K columns

Configuration type	Name	Recommended value
Data Integration Service	INFA_MEMORY	'-Xms1024m -Xmx4096m'
Data Integration Service	maxDTMProcesses	12
Spark session properties	spark.driver.memory	12G
Spark session properties	spark.executor.memory	9G
Tenant specific features	Data Profiling Job limits.memory - Large	9216M

Table 2. 12.5 K tables and 1.3 million columns

Configuration type	Name	Recommended value
Data Integration Service	maxDTMProcesses	12
Data Integration Service	INFA_MEMORY	'-Xms2048m -Xmx4096m'
Data Integration Service	JVMOption1	-Xms2048m
Data Integration Service	JVMOption2	-Xmx4096m
Metadata Platform Service	mps_maxParallelTasks	2
Metadata Platform Service	mps_maxDeploymentBatchSize	50
Spark session properties	spark.driver.memory	12G
Spark session properties	spark.executor.memory	12G
Tenant specific features	Data Profiling Job limits.memory - Large	18432M
Tenant specific features	Data Profiling Job requests.memory - Large	9216M

Table 3. 15 K tables and 1.5 million columns

Configuration type	Name	Recommended value
Data Integration Service	maxDTMProcesses	12
Data Integration Service	INFA_MEMORY	'-Xms2048m -Xmx4096m'

Configuration type	Name	Recommended value
Data Integration Service	JVMOption1	-Xms2048m
Data Integration Service	JVMOption2	-Xmx4096m
Metadata Platform Service	mps_maxParallelTasks	2
Metadata Platform Service	mps_maxDeploymentBatchSize	50
Spark session properties	spark.driver.memory	12G
Spark session properties	spark.executor.memory	15G
Tenant specific features	Data Profiling Job limits.memory - Large	22528M
Tenant specific features	Data Profiling Job requests.memory - Large	9216M

Informatica recommends that you configure filters to profile a maximum of 400 tables.

The profiling run time and throughput vary based on whether the catalog source is created.

The profiling run time and throughput also depends on the following factors:

- Catalog source type
- Volume of content in the tables or files
- Number of tables, rows, and columns
- Column length
- File type
- Average and maximum file size
- Number of unique and duplicate datasets
- Sampling mode
- Source system hardware configuration
- Disk latency
- Network latency between source systems

Sizing recommendations for data profiling tasks in the elastic runtime environment

You can optimize data profiling tasks that use the elastic runtime environment to profile complex file types such as Parquet and Avro in Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2 catalog sources. Configure the size of the advanced cluster based on the total data volume that you want to profile. This reduces the risk of storage and memory failures.

Data profiling tasks that use the elastic runtime environment can fail for the following reasons:

- A worker node runs out of ephemeral storage.
- An executor Java Virtual Machine (JVM) runs out of heap memory.

Sample use case

You want to run a single data profiling task on multiple folders in an Amazon S3 bucket. The folders hold partitioned Parquet files with more than 1 billion rows. If you size the cluster to profile objects in a single folder instead of the combined data volume, the task can run out of storage and fail. To avoid failure and complete the task in the shortest time, size the cluster based on the combined data volume. You can also enable partition detection and pruning to reduce the run time for partitioned Parquet files.

Configure the Secure Agent first, and then size the advanced cluster. The Secure Agent submits data profiling tasks to the advanced cluster and returns the profile results.

Secure Agent configuration

The following table lists the minimum requirements for the Secure Agent:

Resource	Minimum requirement
Disk space	100 GB
Memory	30 GB for data volumes up to 175 million rows, and 64 GB for larger data volumes
CPU cores	16

Size the advanced cluster

An advanced cluster is a Kubernetes cluster that provides a distributed processing environment on the cloud. For data profiling tasks on complex file types, you run the task on a fully-managed or local cluster in an elastic runtime environment.

Create an elastic runtime environment and an advanced configuration in Administrator. Then, on the **Advanced Clusters** page, configure the master instance type, worker instance type, number of worker nodes, and Amazon Elastic Block Store (EBS) volume size.

To size the cluster, complete the following steps:

1. Calculate the total data volume by adding each folder's size, because the data profiling task processes the folders together and stores temporary data on the same worker nodes.
2. Provision an advanced cluster with a master node and worker nodes that use the m5.4xlarge instance type. This instance type has 16 vCPU cores and 64 GB of RAM. Use 1 through 6 worker nodes.
3. Run the data profiling task on the combined folders.

If the task fails with an ephemeral storage error, such as `No space left on device` or `The node was low on resource: ephemeral-storage`, increase the worker nodes to 4 through 10 and increase the EBS volume size for the combined data volume.

Advanced cluster configuration

Configure the following settings on the advanced cluster to tune data profiling performance:

- Partition detection and pruning. When you enable this setting, profiling tasks run faster but require more memory. When you disable it, profiling tasks handle larger data volumes but take longer to complete.
- EBS volume. Set the minimum and maximum EBS volume size for the worker nodes. On an advanced cluster, the volume uses the gp2 volume type. The volume size can range from 50 GB through 16 TB, with a default of 100 GB. Increase the volume size for larger data volumes.
- Executor memory. Increase the executor memory with the `spark.executor.memory` Spark session property. Increase this value when the profiling task runs out of memory on the worker nodes.

Partition detection and pruning enabled

Consider the following configurations when you enable partition detection and pruning:

- **Configuration A**: uses m5.4xlarge instances for the master and worker nodes. Use configuration A for large volumes of data because each worker node runs more executors and the master node has enough memory to schedule and track the partition results.

The following table lists the total Parquet files, total rows, total data elements, worker nodes, and EBS volume per node for configuration A:

Total Parquet Files	Total Rows	Total Data Elements	Worker Nodes	EBS Volume for Each Node
1000	1 billion	151	Minimum worker node required is 1 Maximum worker node required is 6	600 GB
3000	3 billion	453	Minimum worker node required is 4 Maximum worker node required is 10	800 GB

- **Configuration B**: uses an m5.xlarge instance for the master node and m5.2xlarge instance for worker nodes to reach the same parallelism. Use configuration B for smaller data volumes, when you want to use smaller instance types than configuration A, or when a profiling task on configuration A fails with an ephemeral storage error.

The following table lists the total Parquet files, total rows, total data elements, worker nodes, and EBS volume per node for configuration B:

Total Parquet Files	Total Rows	Total Data Elements	Worker Nodes	EBS Volume for Each Node
50	50 million	151	Minimum worker node required is 1 Maximum worker node required is 1	100 GB
60	60 million	151	Minimum worker node required is 1 Maximum worker node required is 3	100 GB
70	70 million	151	Minimum worker node required is 1 Maximum worker node required is 3	100 GB
85	85 million	151	Minimum worker node required is 1 Maximum worker node required is 3	100 GB

Total Parquet Files	Total Rows	Total Data Elements	Worker Nodes	EBS Volume for Each Node
100	100 million	151	Minimum worker node required is 1 Maximum worker node required is 3	100 GB
125	125 million	151	Minimum worker node required is 1 Maximum worker node required is 3	100 GB
150	150 million	151	Minimum worker node required is 1 Maximum worker node required is 3	100 GB

Note: For profiling more than 1 billion rows, upgrade the master node to m5.2xlarge or higher to avoid the out-of-memory failure.

Partition detection and pruning disabled

Use an m5.xlarge instance for the master node and m5.2xlarge instance for worker nodes, and a 100 GB EBS volume, for data volumes up to 1 billion rows.

The following table lists the total Parquet files, total rows, total data elements, number of worker nodes, and EBS volume per node for configurations where partition detection and pruning are disabled:

Total Parquet Files	Total Rows	Total Data Elements	Minimum and Maximum Worker Nodes	EBS Volume for Each Node
175	175 million	151	1	100 GB
200	200 million	151	1	100 GB
250	250 million	151	1	100 GB
300	300 million	151	1	100 GB
400	400 million	151	1	100 GB
500	500 million	151	1	100 GB
600	600 million	151	1	100 GB
700	700 million	151	1	100 GB
800	800 million	151	1	100 GB
900	900 million	151	1	100 GB
1000	1 billion	151	1	100 GB

Estimate the EBS volume size

To profile objects in multiple folders in one run, add the rows across all folders and use the combined value.

The following table lists row ranges and the recommended EBS volume size for Parquet files when you enable partition detection and pruning:

Row Range	EBS Volume for Each Node
Up to 150 million	100 GB
150 million through 350 million	200 GB
350 million through 600 million	400 GB
600 million through 1.2 billion	600 GB
1.2 billion through 2 billion	1 TB
2 billion through 3 billion	1.5 TB

Tuning guidance

- Size the cluster for the total data volume that you profile, not for the row count.
- Enable partition detection and pruning for faster run times.
- Disable partition detection and pruning to profile large data volumes. This setting is more sensitive to memory, so large data volumes can cause an out-of-memory failure when it is enabled.
- Organize the source into many smaller files rather than a few large files to improve reliability.

Increase the Java heap size

An insufficient Java heap size can result in metadata load failures. To avoid such issues, you can incease the heapSize parameter value of the Metadata Platform Services or Metadata Foundation Application services.

1. Log in to Informatica Intelligent Cloud Services and open Administrator.
2. Click **Runtime Environments** and select the Secure Agent for which you want to increase the heapSize parameter value.
3. Click **Edit**.
4. Go to the **Custom Configuration Details** section.
5. Select the **Metadata Foundation Application** or **Metadata Platform Services** service.
6. Select the following type: **PLUGIN_CFG**.
7. Enter the following name: `maxHeap_size`
8. Enter the value, for example 4096 MB or 8192 MB.

On a source system with 15 K tables and 1.5 million columns, when you apply a data quality rule to a column, it's recommended that you create a maximum of 8 K rule occurrences with Agent Cache and 6 K rule occurrences with No Cache.

Author

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